جامعة الأمير سلطان
PRINCE SULTAN UNIVERSITY

{CS} Department of Computer Science

CS 499 Senior Project

CCIS كلية علوم الحاسب والمعلومات COLLEGE OF COMPUTER & INFORMATION SCIENCES

Form B Senior Project Proposal

Project Title: **LLM Data Scientist**	
Advisor: **Dr. Omar Alomeir**	
Student Names: **Ali Ibrahim-222110939 / Elsayed Azab-221110389 / Mohammed Sharaf-222110626**	

Guideline:
There are several factors that make a suitable senior project:

- A senior project should have a well-defined beginning and end of the project as well as clearly-defined criteria for success or failure.
- The student needs to be the sole responsible for the project. This means that the student is directly responsible for the success or failure of the project. Any risk factors in the project should be thoroughly evaluated including unforeseen circumstances that cause failures, etc.
- The project requires the student to do some investigation, thoroughly analyze the requirements, complete the design before any implementation
- Students would conduct literature review, and see how others approached the selected problem? What new technologies will be involved? What new techniques or technologies must the student learn (or invent) to successfully complete this work?
- The project should require creativity on the part of the student. The solution should not be obvious and must be the student's own work.
- The student(s) may work in a team of up to 4 students. In this case, students' work quality and quantity should be of reasonable proportions that would be approved by the course supervisor.

Once you or your team has addressed the above-mentioned factors, you will provide a project proposal using the following sections as guideline:

- Problem Statement
- Project Objectives
- Earlier/Similar Work/App
- Project tasks and task assignment (Demonstrate individual and Group Contribution):
    - Break down the work into smaller tasks and phases.
    - Assign each task to a specific team member.
    - Ensure responsibilities are well-distributed and clearly defined.
- Detailed Project execution plan with timely milestones (stages) for CS499
    - Gantt/roadmap for remaining weeks; team responsibilities.
- Expected deliverables for each stage

- **Problem Statement**
  Large Language Models (LLMs) are widely used to assist with data analysis, but it is still unclear how reliable they are as "data analysts" when the task requires correct statistical reasoning, proper handling of missing data, and especially survey methodology (weights, design effects, variance estimation, cross-country comparisons).

  This project will evaluate whether modern LLMs can generate correct descriptive + inferential statistical analyses on real, large-scale survey datasets, and whether agentic architectures (single-agent, multi-agent, and retrieval-augmented generation RAG) improve reliability. The outputs will be benchmarked against gold-standard expert analyses.

- **Project Objectives**
  1. Build a controlled evaluation pipeline to test LLMs as data analysts on real survey datasets.
  2. Produce expert "ground-truth" analyses (descriptive + inferential) for selected research questions.
  3. Test multiple LLMs (and/or configurations) on the same tasks using standardized prompts.
  4. Compare architectures:
     - **Single-agent** (one-shot or iterative prompting)
     - **Multi-agent** (roles like planner/statistician/reviewer)
     - **RAG agent** (retrieves documentation/code snippets/rules for survey stats)
  5. Quantify performance:
     - statistical correctness (method choice, results, interpretation)
     - survey methodology correctness (weights/design/variance)
     - clarity and reproducibility of the report
  6. Deliver final report + documented code + presentation of findings.

- **Earlier/Similar Work/App**
  Prior studies explored LLMs in survey analysis and statistical reasoning, but reliability remains uncertain—especially with complex, real-world survey designs. Related work includes research on LLMs for coding survey responses and evaluating statistical competence, and recent systems for automated data science using agentic LLMs (e.g., "DeepAnalyze/DeepAnalyzer" style agent workflows). Our project extends this by **benchmarking LLM-generated survey analysis outputs against expert ground truth** and explicitly testing whether **agentic designs (multi-agent/RAG)** improve accuracy.

- **Project tasks and task assignment** (Demonstrate individual and Group Contribution):

  Phase A: Planning & Literature

  - [x] ~~Define scope, datasets, and evaluation criteria~~
  - [x] ~~Literature review (LLMs + survey statistics + agentic data science)~~

  Phase B: Data Preparation

  - [x] ~~Acquire datasets (e.g., WVS Wave 7, Arab Barometer, GSS)~~
  - [x] ~~Cleaning, missing-data handling plan~~
  - [ ] Define subset variables and research questions

  Phase C: Ground Truth Analysis (Expert Baseline)

  - [ ] Implement correct analyses in R/Python (with survey weights/design)
  - [ ] Produce “gold standard” tables + interpretation write-ups

  Phase D: LLM Experiments

  - [ ] Prompt design and standardization
  - [ ] Run LLMs on tasks (descriptive + inferential)
  - [ ] Collect outputs and logs

  Phase E: Agentic Architectures

  - [ ] Multi-agent workflow design (planner/statistician/checker)
  - [ ] RAG setup (retrieval of survey-stat rules + code templates)
  - [ ] Run same tasks under each architecture

  Phase F: Evaluation & Reporting

  - [ ] Compare against ground truth
  - [ ] Score errors (method mistakes, weight mistakes, interpretation mistakes)
  - [ ] Final report + slides + repo

- Detailed Project execution plan with timely milestones (stages) for CS499
    - Gantt/roadmap for remaining weeks; team responsibilities.
- Expected deliverables for each stage